

Knowledge Graphs: A Method for Visually Representing Technology- induced Errors to Reason About Digital Health Safety Issues

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Introduction: Technology-induced Errors

- **Digital Systems of Care (that include AI)**
 - can *improve care processes* while at the *same time introducing new types of errors* that have significant impacts for human (patient, health professional stakeholders).
 - With the *exponential rise in the use of AI applications and systems* in varying health settings (i.e., hospital, community, and home), there is an even *greater need to ensure our digital health ecosystems of care are safe*.
- **Managing Health Technology Risk**
 - is an important part of *technology design, implementation and evaluation* in a modern digital, healthcare system that uses AI to support clinical decision making.

Objectives

1. ***Review*** the evolution of the field of ***technology safety in healthcare***.
2. Discuss some of the ***issues and considerations*** associated with ***health technology risk management***.
3. Methods will be presented that can be used to ***assess and ensure safety*** in health environments as part of a healthcare organizational strategy for ***using and improving AI***.
4. Examine the ***role of Knowledge Graphs to support this process***.
5. Examine new approaches to ***validating Knowledge Graphs***.

What are Technology-induced Errors?

Technology-induced errors are errors that arise from the:

- a. Design and development of a technology
- b. Its implementation and customization
- c. Interactions between the operation of the technology and the new work processes that arise from a technology's use
- d. Its maintenance
- e. The interfacing of two or more technologies used to provide or support healthcare services.

Borycki EM, Kushniruk AW, Bellwood P, Brender J. Technology-induced errors. The current use of frameworks and models from the biomedical and life sciences literatures. *Methods Inf Med*. 2012;51(2):95-103. doi: 10.3414/ME11-02-0009. Epub 2011 Nov 21. PMID: 22101488.

Kushniruk A, Surich J, Borycki E. Detecting and classifying technology-induced error in the transmission of healthcare data. In: 24th International Conference of the European Federation for Medical Informatics Quality of Life Quality of Information 2012 (Vol. 26).

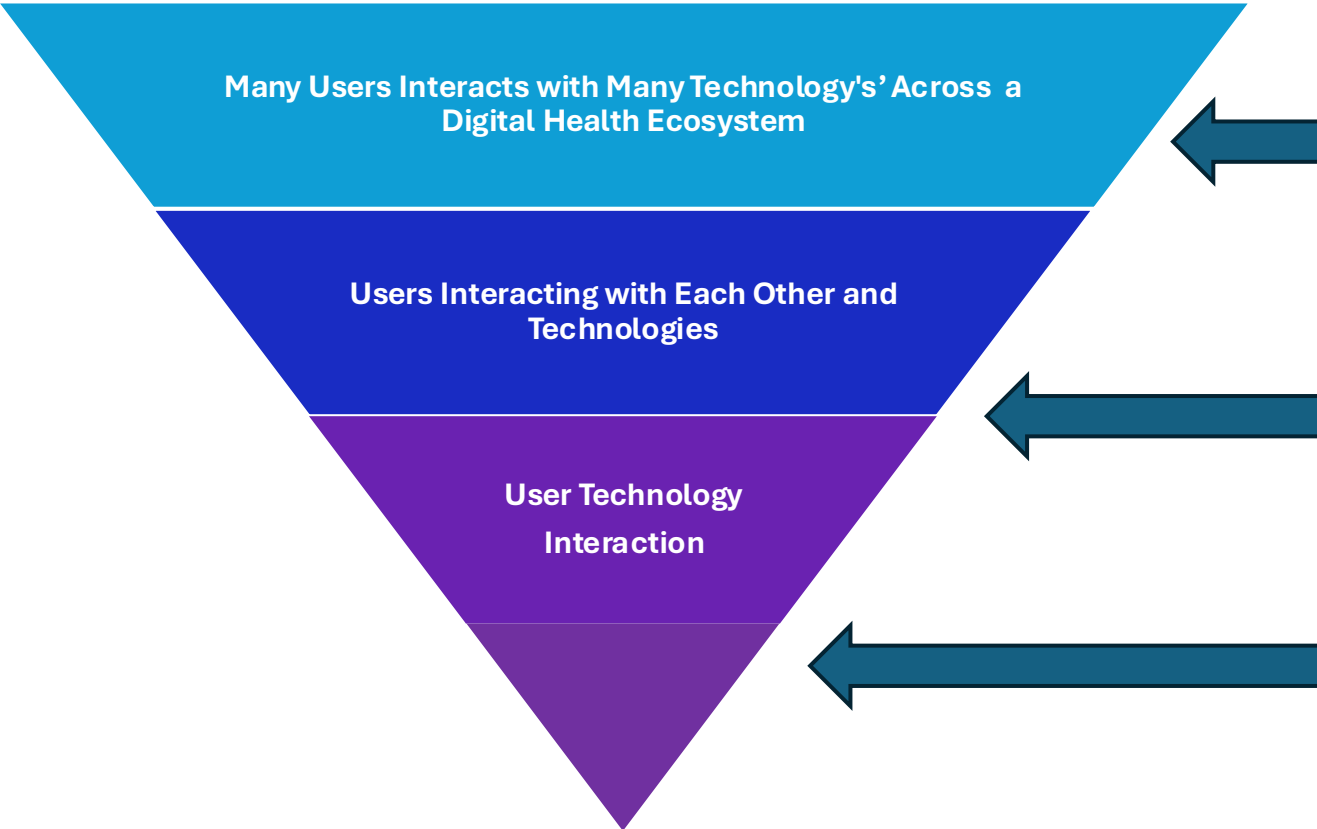
Kushniruk AW, Triola MM, Borycki EM, Stein B, Kannry JL. Technology induced error and usability: the relationship between usability problems and prescription errors when using a handheld application. *Int J Med Inform*. 2005 Aug;74(7-8):519-26. doi: 10.1016/j.ijmedinf.2005.01.003. Epub 2005 Apr 8. PMID: 16043081.

Examples of Technology-induced Errors

Table 2. Examples of methods and tools focused on technology, digital ecosystem, and digital infrastructure safety.

Examples	References
Methods to assessing and evaluating safety during health technology procurements (i.e. evidence-based heuristics, usability testing, clinical simulations, computer based simulations)	29,33-36,39,40,42,43
Models and frameworks for understanding technology-induced errors	30,34
Design checklists for software user interface safety	38-40
Design guidelines that improve software user interface safety	38-40
Approaches that test and evaluate the safety of user interfaces and their connection to usability and workflow	10,11,44,45
Methods for safety testing user interfaces, workflows and digital integrations for quality and safety before, during and after systems implementation (i.e. heuristics, usability testing, clinical simulation, computer based simulation)	10,11,29,33-36,39,40,42-45
Software configuration checklists for safety	38-40
Dashboards for viewing quality and safety issues	46
Incident reporting	8,9
Taxonomies specific to technology-induced errors (data and text analytics)	7-9,47
Manual and automated approaches to analyzing incident reports and detecting errors	7-9,48-50
Incident investigation to improve systems	51
Employee digital safety surveys	52
Downtime and technology failure management	53,54
Learning from reports and recalls	55

How Do We Make Technology Safer in Healthcare?



Many Users Interacts with Many Technology's' Across a Digital Health Ecosystem

Organizational Level

Users Interacting with Each Other and Technologies

Workflow Level

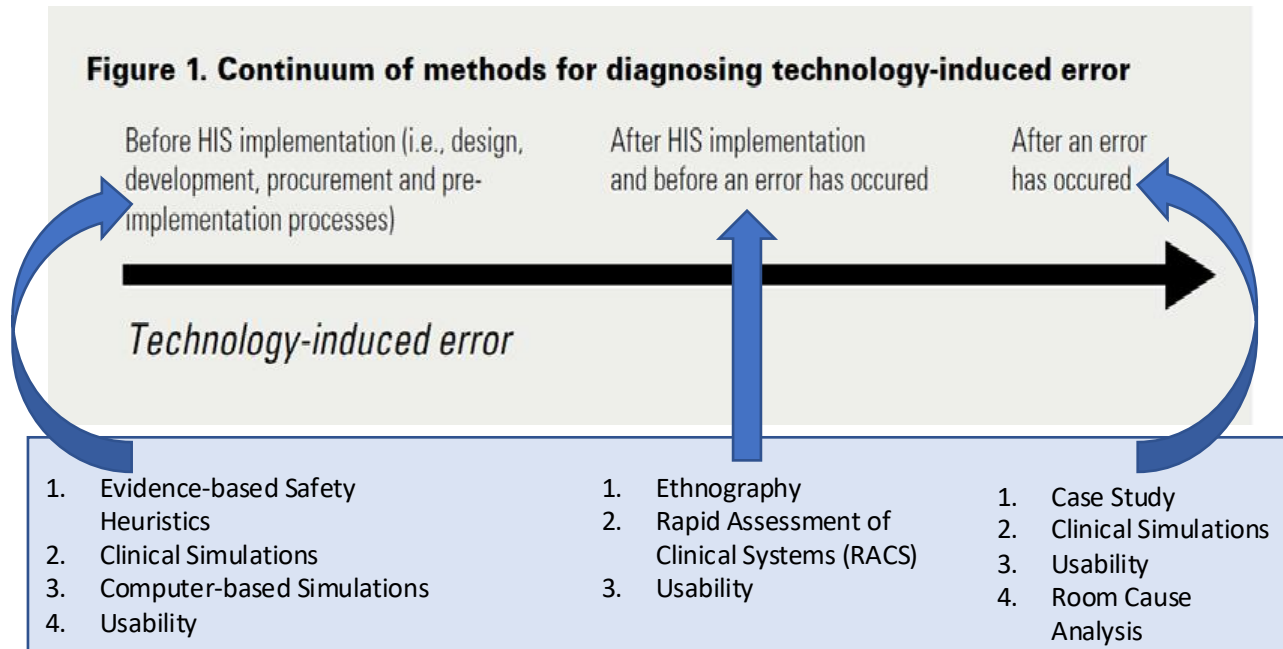
User Technology Interaction

Human-computer Interaction Level

Cognitive Socio-Technical Theory: A Way to Diagnose Technology-induced Errors

Borycki EM, Kushniruk AW. Towards an integrative cognitive-socio-technical approach in health informatics: analyzing technology-induced error involving health information systems to improve patient safety. The open medical informatics journal. 2010 Sep 15;4:181.

Methods for Diagnosing Technology-induced Errors



Borycki E, Keay E. Methods to assess the safety of health information systems. *Healthc Q*. 2010;13 Spec No:47-52. doi:10.12927/hcq.2010.21966

Borycki, E., Dexheimer, J. W., Hullin Lucay Cossio, C., Gong, Y., Jensen, S., Kaipio, J., Kennebeck, S., Kirkendall, E., Kushmiruk, A. W., Kuziemy, C., Marcilly, R., Röhrig, R., Saranto, K., Senathirajah, Y., Weber, J., & Takeda, H. (2016). Methods for Addressing Technology-induced Errors: The Current State. *Yearbook of medical informatics*, (1), 30–40. <https://doi.org/10.15265/IY-2016-029>

Clinical Simulations: Visualizing the Impact of Technology-induced Errors Over Time



Study of Relationship of Usability to Technology-induced Error

- Evaluation of a commonly used mobile phone ePrescribing application
- **Study Objectives:**
 - To assess if the application introduced errors
 - To provide feedback into redesign of user interface
 - To explore the relationship between usability problems and medical error (specifically medication data entry errors)

Materials

- **Software**

- Mobile phone ePrescribing software
- Contains a database of 8,000 medications
- Allows physician users enter and store medications, print and process them

- **Hardware**

- Mobile phone
- Connected to a data projector for viewing of mobile phone ePrescribing software user interface

Methods

- **Subjects**
 - 10 physicians
 - Mobile phone users who had not used the ePrescribing software being used
- **Procedure**
 - Each subject received training on use of the software
 - Subjects were then asked to:
 - Enter medications from a paper list (as accurately as possible)
 - Read a clinical scenario involving patient cases and enter medications
 - Subjects were asked to “think aloud”
 - All mobile phone screens were recorded

Analysis

1. Physicians' think aloud verbalizations were audio recorded.
2. Physicians' interactions with the ePrescribing system were video recorded.
3. Data are qualitatively analyzed using an evidence-based coding scheme for usability problems and errors by two independent coders.

Borycki E, Kushniruk A. Identifying and preventing technology-induced error using simulations: application of usability engineering techniques. *Healthc Q*. 2005;8 Spec No:99-105. doi: 10.12927/hcq..17673. PMID: 16334081.

Kushniruk AW, Triola MM, **Borycki EM**, Stein B, Kannry JL. Technology induced error and usability: the relationship between usability problems and prescription errors when using a handheld application. *International journal of medical informatics*. 2005 Aug 1;74(7-8):519-26.

Evidence-based Coding Scheme for Usability Problems and Errors



Cognitive Activities

- Clinical reasoning
- Problem solving
- Decision making
- Sensemaking
- Hypothesis generation
- Information integration



Human-Computer Interaction

- Clicking
- Navigation errors
- Information searching
- Data entry problems
- Alert interactions



Socio-Technical

- Workflow interruptions
- Communication breakdowns
- Coordination issues
- Organizational constraints
- Environmental influences
- Team interactions



Safety/ Technology-induced Errors

- Near misses
- Errors
- Information omissions
- Misinterpretations
- Unsafe workarounds

Usability Problems and Medication Entry Errors

Table 1 Usability problems and their relationship to medication entry errors.

		Subjects										Total usability problems by category	Total errors	% Usability problem associated with error
		S1	S2	S3	S4	S5	S6	S7	S8	S9	S10			
Problem														
Interface														
	Data entry		1	3		1	1				3	9	7	77.8
	Display visibility	1	3	1	2	2		1	1	5	3	19	16	84.2
	Navigation				1	1			1	1		4		0.0
	Locating	1					2			3		6	1	16.7
	Procedure	1		1	2	1	1	1	1	2	1	11		0.0
	Printing		2	1		1	2			1	1	8	1	12.5
	Speed								1	1	1	3		0.0
	Attention								1			1		0.0
	Total	3	6	6	5	6	6	2	5	13	9	61	25	41.0
Content														
	Database	4	1	1							2	8		0.0
	Default	1								1	1	3	2	66.7
	Training manual									1		1		0.0
	Total	5	1	1	0	0	0	0	0	2	3	12	2	16.7
Grand total		8	7	7	5	6	6	2	5	15	12	73	27	37

Kushniruk AW, Triola MM, Borycki EM, Stein B, Kannry JL. Technology induced error and usability: the relationship between usability problems and prescription errors when using a handheld application. Int J Med Inform. 2005 Aug;74(7-8):519-26. doi: 10.1016/j.ijmedinf.2005.01.003. Epub 2005 Apr 8. PMID: 16043081.

Probability of Errors

Table 2 Errors broken down into slips and mistakes.

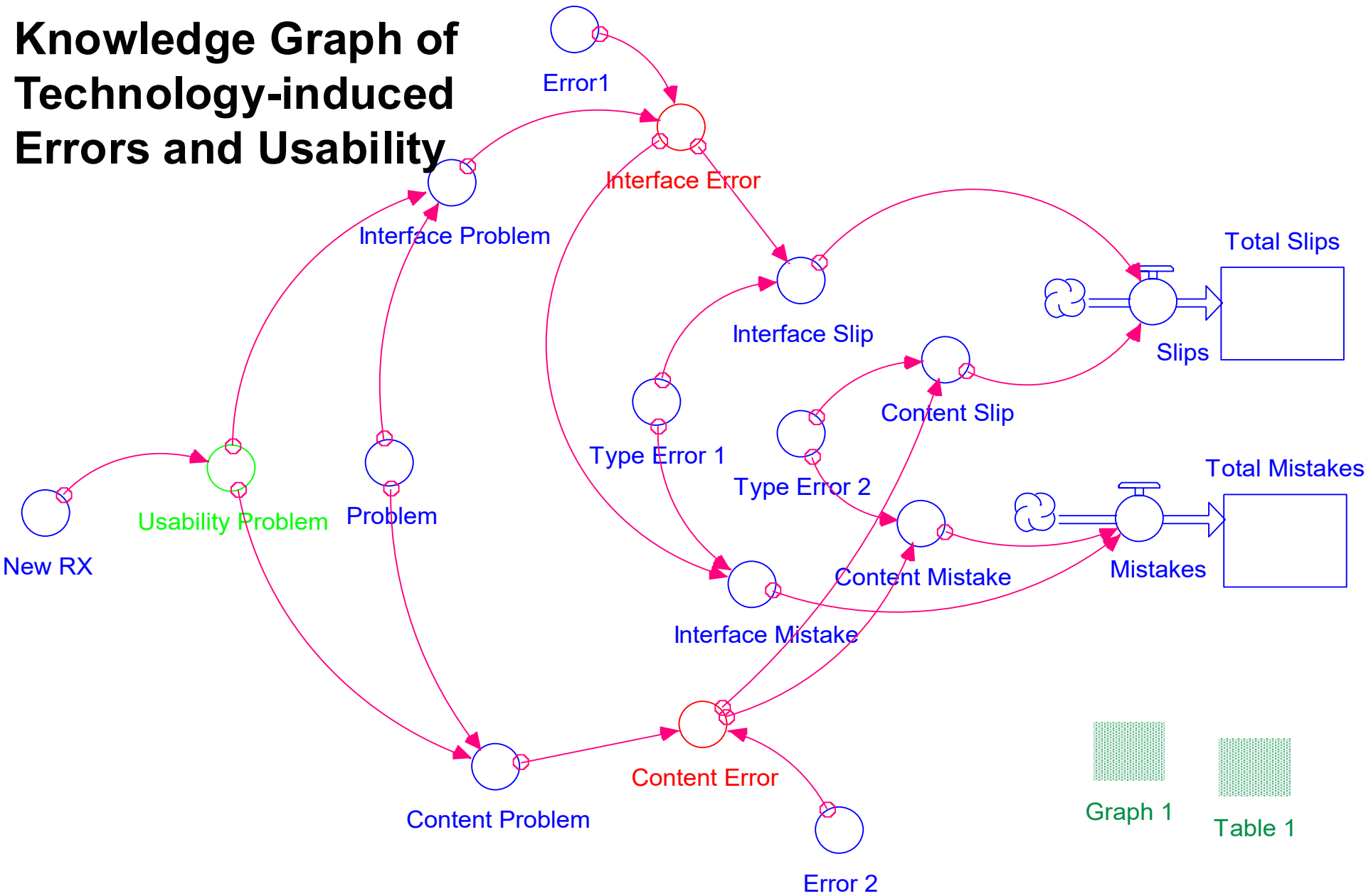
	Total usability problems by category	Slips	Mistakes	Total errors	% Usability problem associated with error	Probability of an error occurring	Odds of an error occurring
Problem Interface							
Data entry	9	7		7	77.8	0.8	4:1
Display visibility	19	4	12	16	84.2	0.8	4:1
Navigation	4				0.0	0	0
Locating	6	1		1	16.7	0.2	1:4
Procedure	11				0.0	0	0
Printing	8	1		1	12.5	0.1	1:9
Speed	3				0.0	0	0
Attention	1				0.0	0	0
Total	61	13	12	25	41.0	0.4	2:3
Content							
Database	8				0.0	0	0
Default	3	1	1	2	66.7	0.7	3:7
Training manual	1				0.0	0	0
Total	12	1	1	2	16.7	0.2	1:4
Grand total	73	14	13	27	37	0.4	2:3

Kushniruk AW, Triola MM, Borycki EM, Stein B, Kannry JL. Technology induced error and usability: the relationship between usability problems and prescription errors when using a handheld application. Int J Med Inform. 2005 Aug;74(7-8):519-26. doi: 10.1016/j.ijmedinf.2005.01.003. Epub 2005 Apr 8. PMID: 16043081.

Input into Computer Simulation (Kushniruk, Borycki, Anderson & Anderson, 2008) from study of users entering prescriptions into handheld application

Parameter	Value
New RX	Random Number (0-1)
Usability Problem	Probability = 1.00
Interface Problem	Probability = 0.84
Content Problem	Probability = 0.16
Interface Error	Probability = 0.41
Content Error	Probability = 0.167
Interface Slip	Probability = 0.52
Interface Mistake	Probability = 0.48
Content Slip	Probability = 0.50
Content Mistake	Probability = 0.50

Knowledge Graph of Technology-induced Errors and Usability



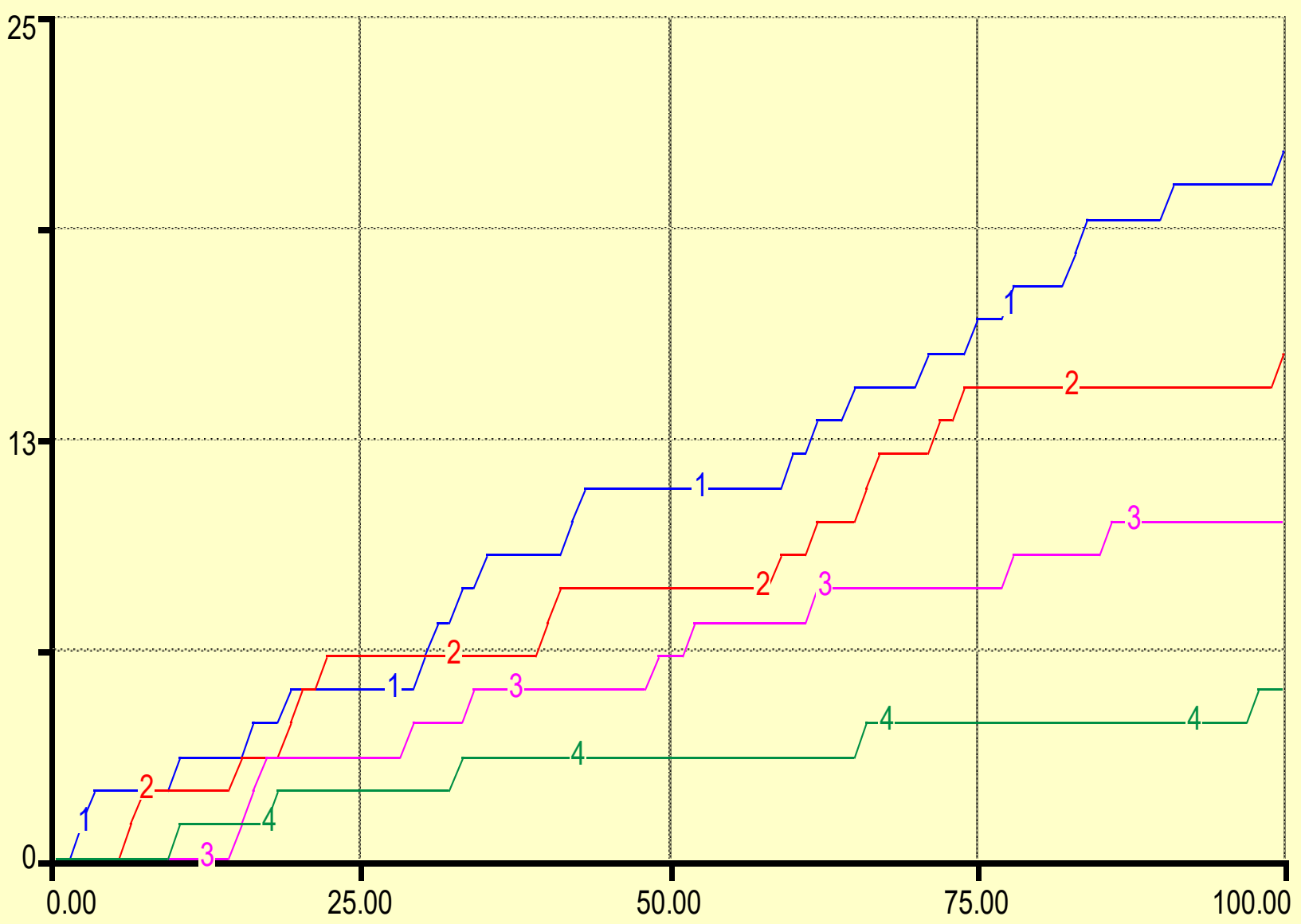
Simulation Runs

Runs	Probability of Interface Problem	Probability of Content Problem
Run 1	0.84	0.16
Run 2	0.60	0.16
Run 3	0.40	0.16
Run 4	0.20	0.16

1:

1:

1:



Comparative Graph of Total Mistakes

Extrapolating Out

- “1 in 3 US physicians use this product to help them reduce medical errors.....”
- Based on clinical simulation study, and assuming 674,824 US Physicians and errors that would occur during learning period alone
 - Over 500,000 errors per year estimated from this application alone
 - New project estimating cost of such error from medical point of view

Analyzing Incident Report Data



Incidents Involving Technology

Individual Practitioner Reporting

Medication Incidents Including Near Misses

Reporting by individual practitioners is an important component in identifying early opportunities for enhancing medication safety. ISMP Canada sincerely appreciates your willingness to share information about medication incidents and related findings. Please include as much information about the incident as possible. **INFORMATION SUBMITTED WILL BE KEPT STRICTLY CONFIDENTIAL AND PROTECTED.** Please do not supply identifying information (e.g., patient name or date of birth, hospital name, or healthcare provider names).

1*	2*	3*	4	5	6*	7
Incident	Outcome	Medication(s)	Follow Up	Patient	Reporter	Contributing Factors

* Indicates Required Fields (Total of nine required fields.)

Date of Incident:

Time of Incident:

Incident Description / How Discovered:*

Stages Involved:*

☐ Prescribing ☐ Order Entry/Transcription
☐ Dispensing/Delivery ☐ Administration
☐ Monitoring ☐ Other

Type of Incident:*

Discovered By:*

Care Area Type:*

protected by reCAPTCHA

- Description of the Incident
- Outcome of the Incident
- Medications
- Follow Up
- Patient Information
- Reporter
- Contributing Factors

<https://ismpcanada.ca/>

Individual practitioner reports of medication incidents contribute to the Canadian Medication Incident Reporting and Prevention System (CMIRPS). Through CMIRPS, some of Canada's most highly regarded organizations committed to patient safety are working together to support your efforts to enhance the safety of the medication system. The collaborating parties of the CMIRPS program are: Health Canada, the Canadian Institute for Health Information (CIHI), the Institute for Safe Medication Practices Canada (ISMP Canada) and the Canadian Patient Safety Institute (CPSI). [More information about ISMP Canada's activities for CMIRPS.](#)

Data Driven Safety: Incident Reports

MAUDE FDA Data

- 2 years of data (2019-2020)
- From the United States (US)
- Focus US EHR Vendors
- Reports focused on technology-induced errors (TIE's)

- Initially coded with concepts from the literature on TIE's
- Standardize the coding scheme
- Auto-code using the coding scheme (Supervised Machine Learning Analyses)
- Observe for TIE across units, organizations and healthcare settings

Implications

- Can be replicated in other settings
- Evolutionary:
 - as new technologies are implemented new errors emerge
- opportunity for continuous learning

Borycki EM, Farghali A, Kushniruk AW. Complexity and Health Technology Safety. Stud Health Technol Inform. 2022 Jun 29;295:551-554. doi: 10.3233/SHTI220787. PMID: 35773933.

Borycki EM, Farghali A, Kushniruk AW. Complexity and Health Technology Safety. Stud Health Technol Inform. 2022 Jun 29;295:551-554. doi: 10.3233/SHTI220787. PMID: 35773933.

Borycki EM, Farghali A, Kushniruk AW. Do Health Technology Safety Issues Vary by Vendor? Stud Health Technol Inform. 2022 Jun 29;295:345-349. doi: 10.3233/SHTI220734. PMID: 35773880.

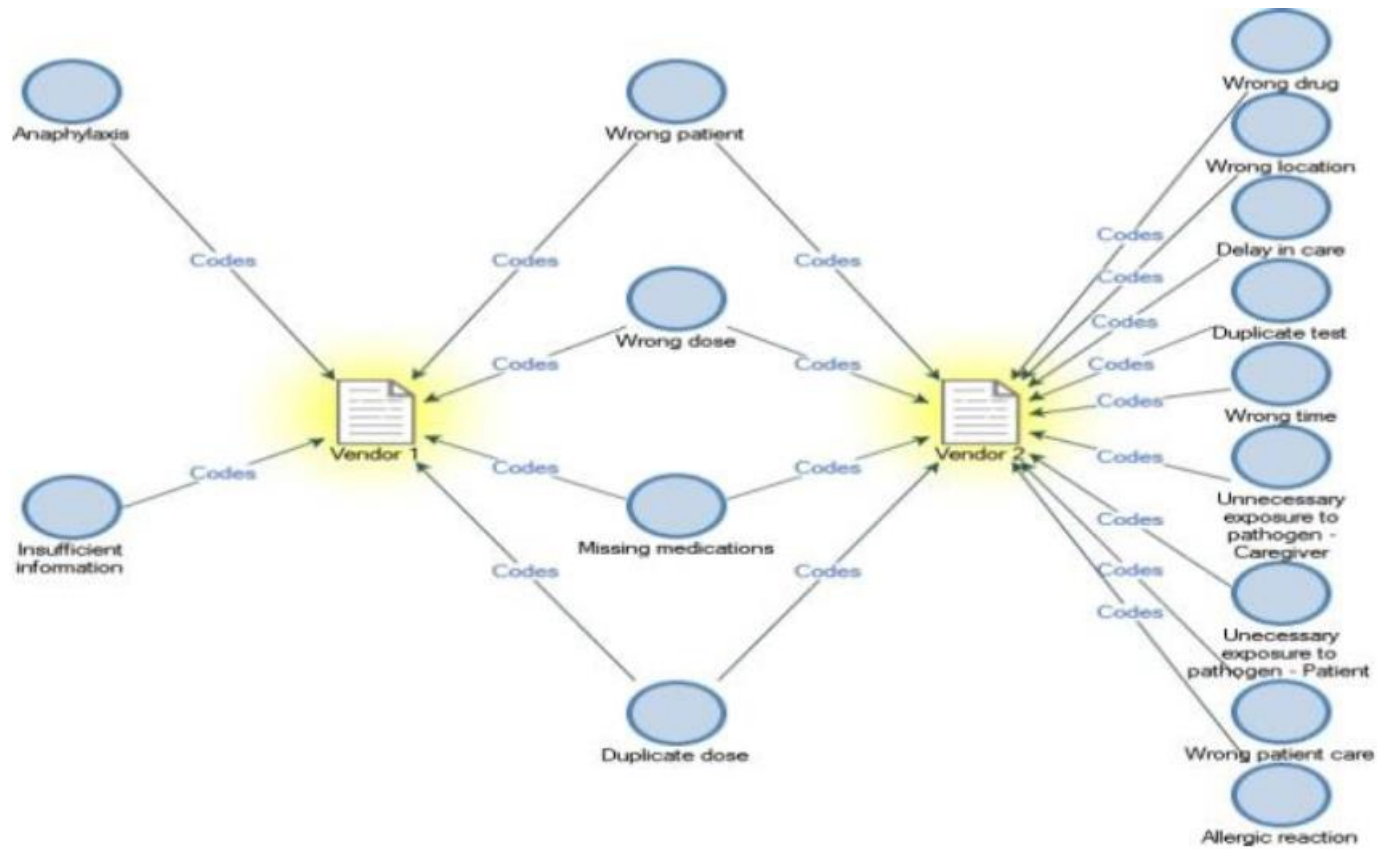


Figure 1: Common Types of Errors

Borycki EM, Farghali A, Kushniruk AW. Do Health Technology Safety Issues Vary by Vendor? *Stud Health Technol Inform.* 2022 Jun 29;295:345-349. doi: 10.3233/SHTI220734. PMID: 35773880.

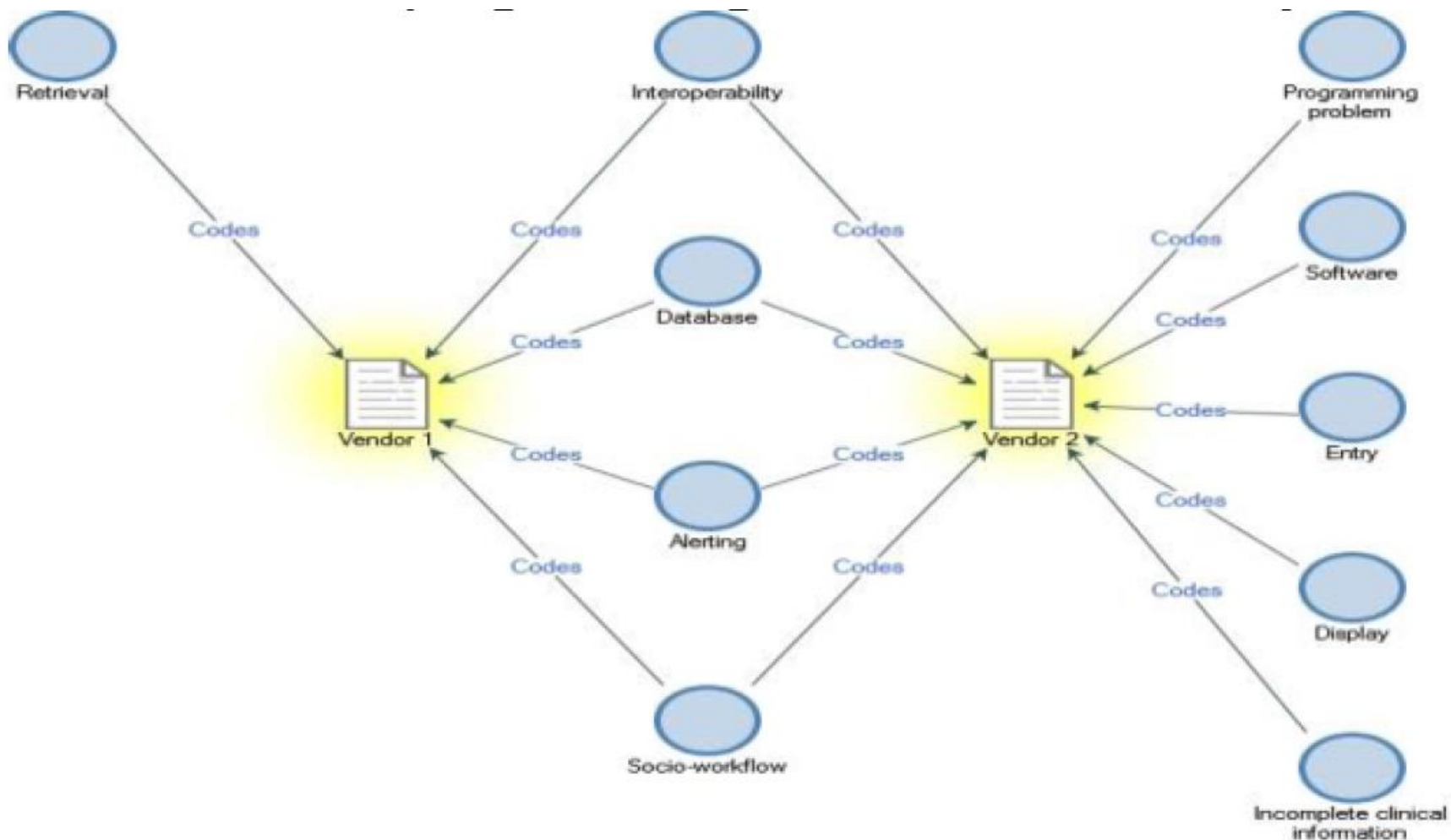


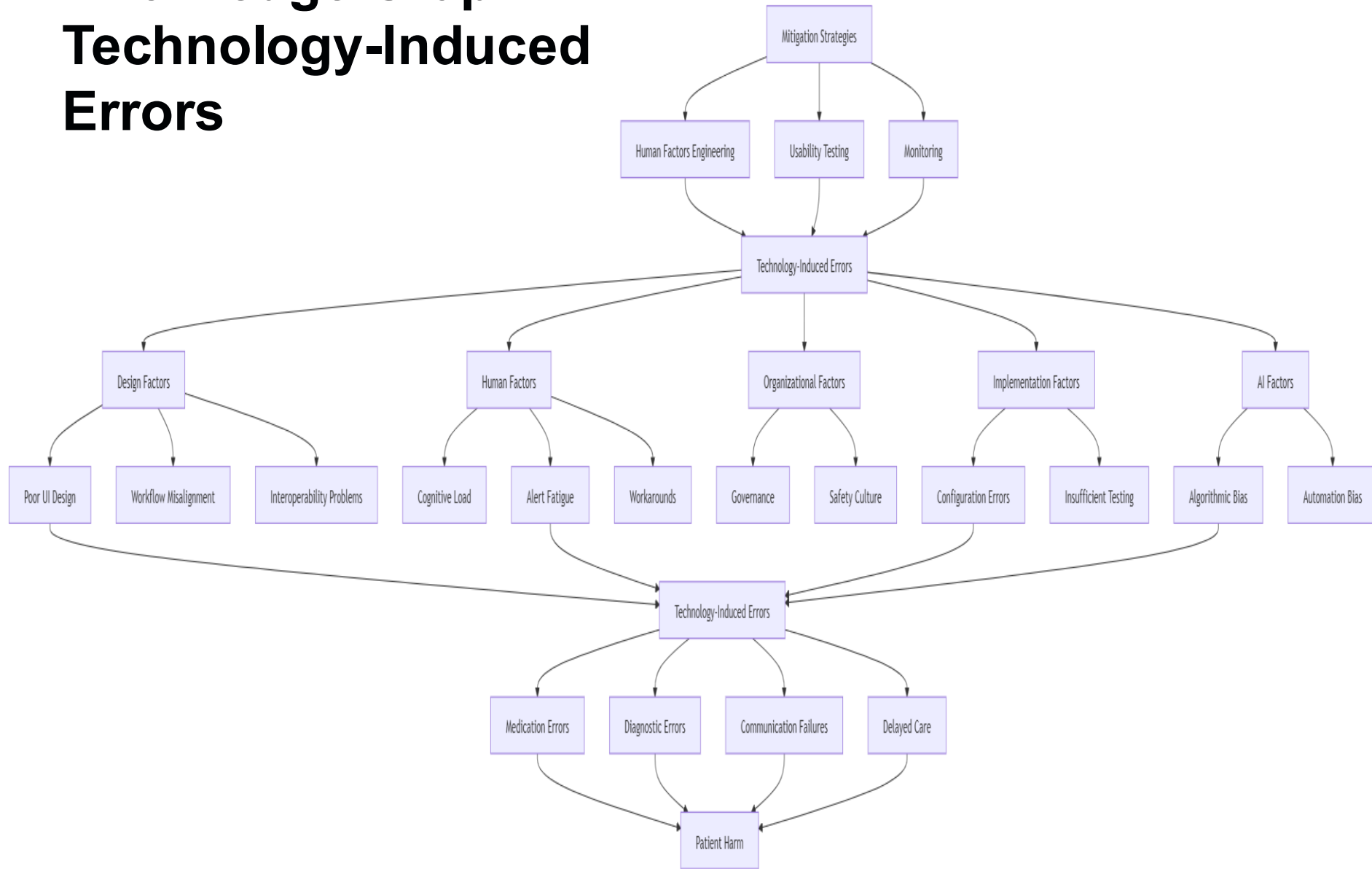
Figure 2: Common Causes of Errors

Borycki EM, Farghali A, Kushniruk AW. Do Health Technology Safety Issues Vary by Vendor? *Stud Health Technol Inform.* 2022 Jun 29;295:345-349. doi: 10.3233/SHTI220734. PMID: 35773880.

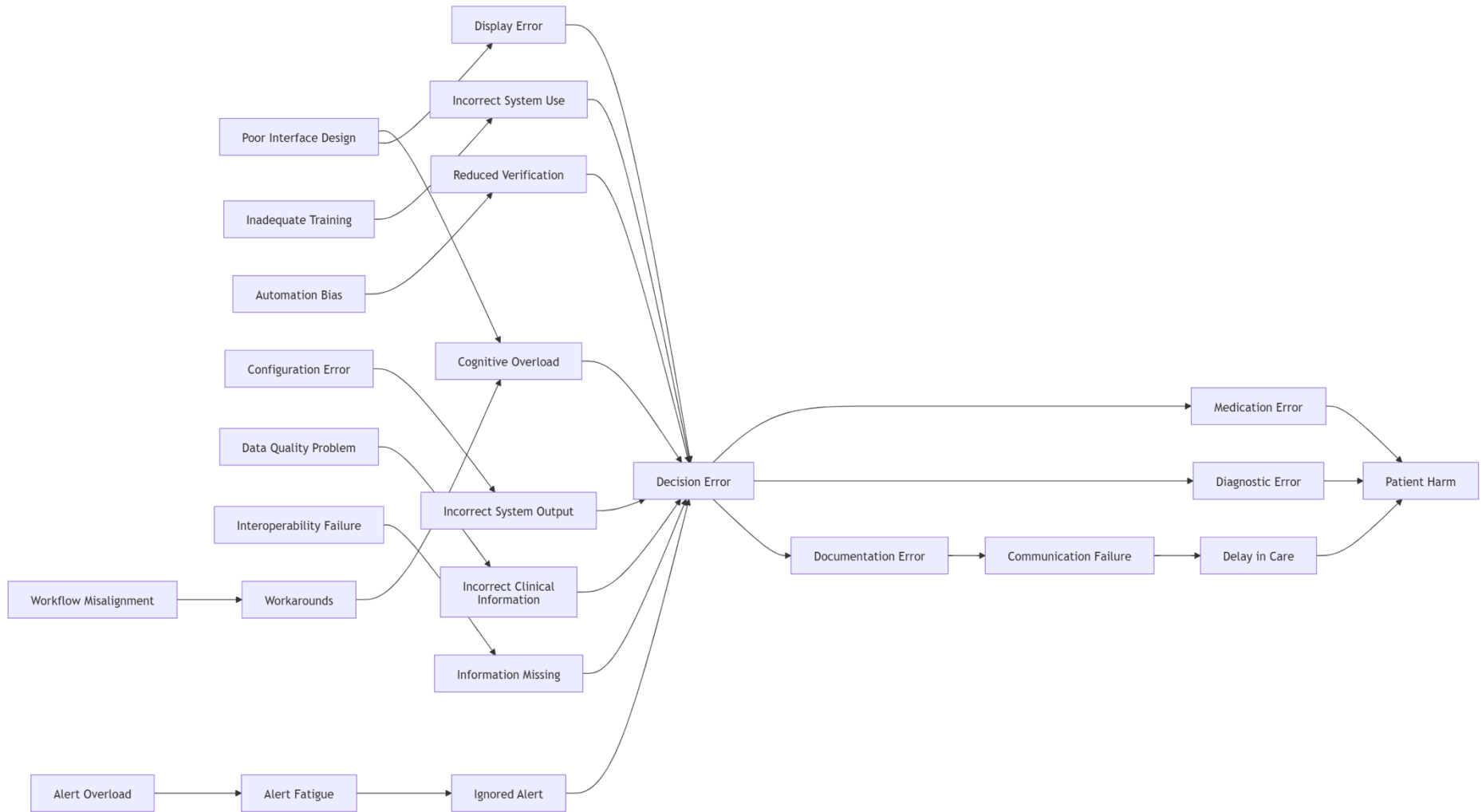
Knowledge Graphs and Technology-Induced Errors



Knowledge Graph: Technology-Induced Errors



Causal Knowledge Graph for Technology-Induced Error Analysis



Knowledge Graphs: Barriers and Challenges

- Identification of the most relevant paths to users reasoning about safety issues
- Removing noise and irrelevant nodes identify the root causes of safety issues
- Missing safety data
- Diversity of the safety data
- Complexity of the safety data
- **Validating knowledge graph quality in representing safety data**

Validating Knowledge Graphs with Users in Laboratory Experiments

1



2



<https://www.flaticon.com/free-icons/ab-testing>

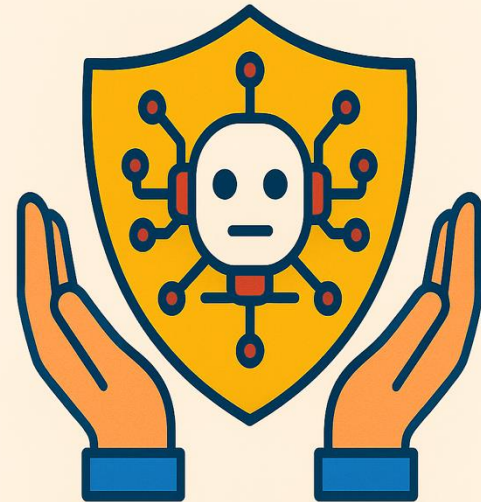
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Validating Knowledge Graphs with Users Using Clinical Simulations

1



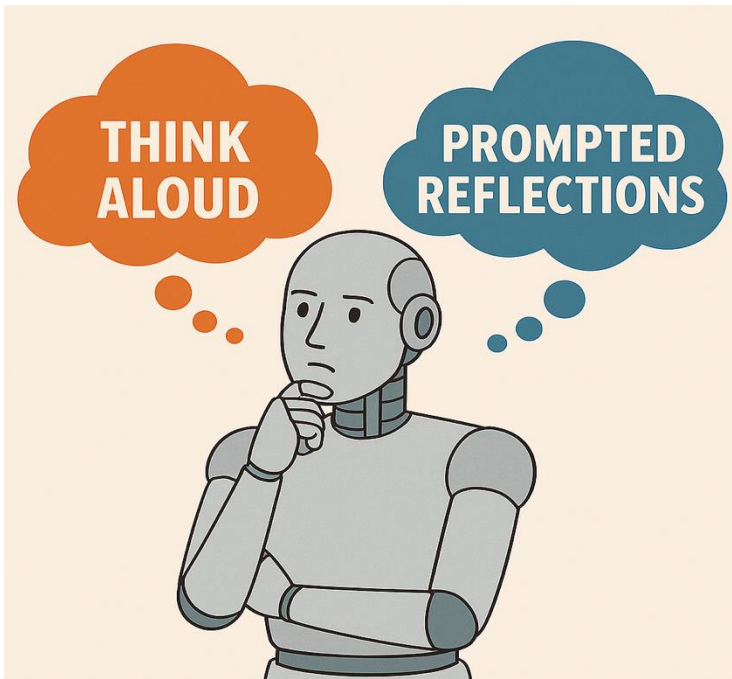
2



AI SAFETY

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Validating Knowledge Graphs in Context



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Thank you!!!



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Global Lab for Digital Health and AI Innovation

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